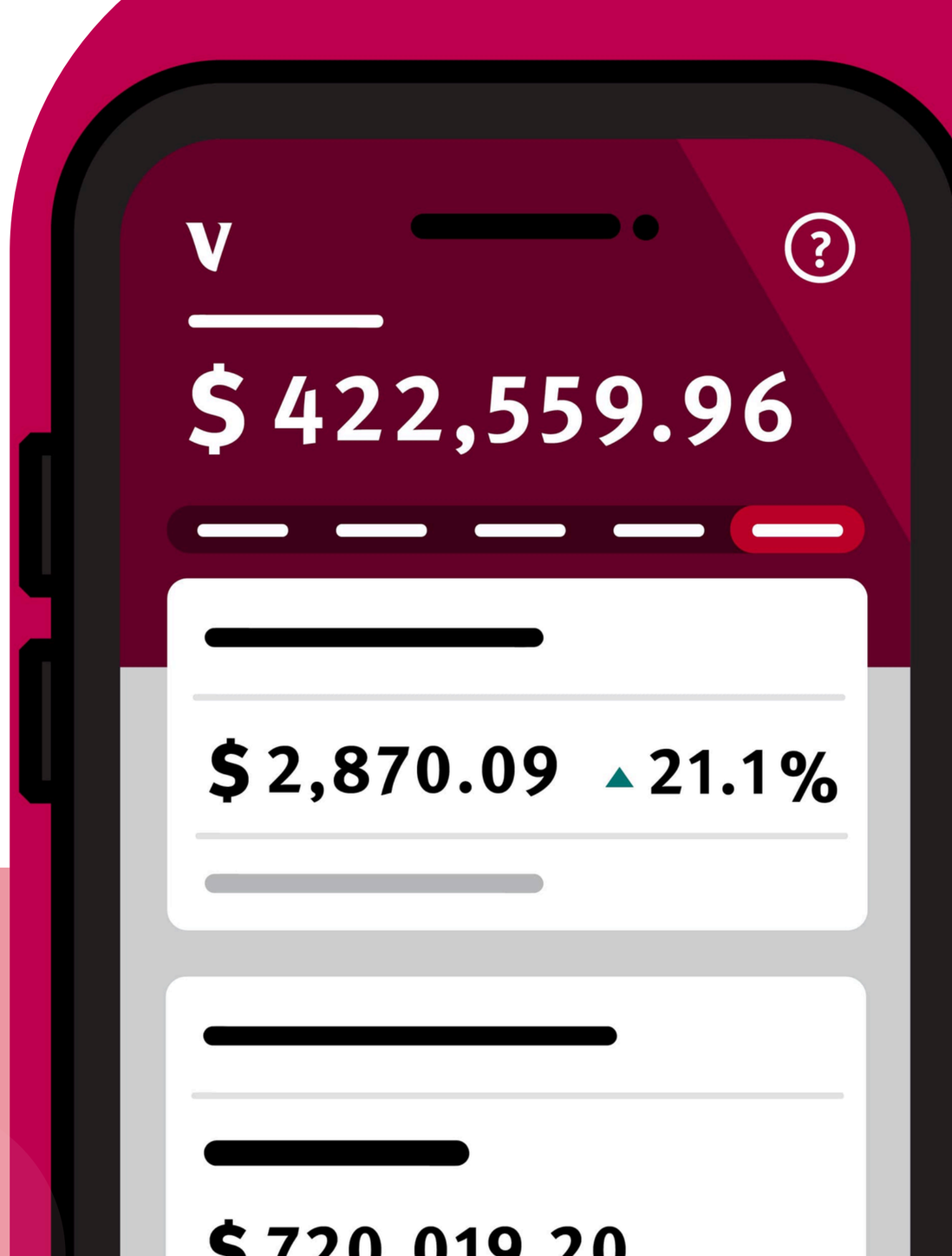
Case Study

EVALUATING VANGUARD'S NEW UI: IMPACT ON USER BEHAVIOR AND CONVERSION



Team Members:

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Project Overview

Vanguard — Company Overview
Vanguard is one of the world's largest investment management companies, offering services such as mutual funds, ETFs, retirement planning, and wealth management.

What sets it apart is its client-first structure. The company is owned by its investors, allowing it to focus on low costs, long-term value, and trust.

Context for this Project
As Vanguard continues to improve its digital experience, this project analyzes an A/B test to evaluate whether a new, more intuitive interface helps clients complete a key process more effectively.

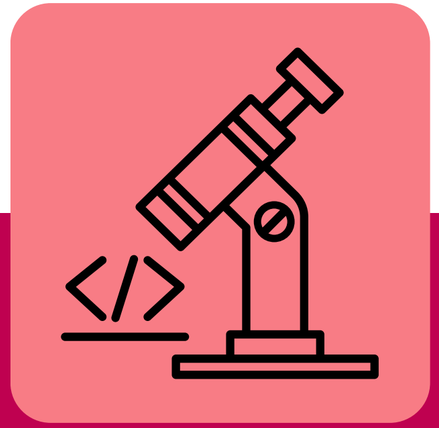
The Three Data Sets

Client Profiles (df_final_demo)

- Age, gender, tenure, accounts, balance

Digital Footprints (df_final_web_data)

- User actions across steps
- Timestamps and session tracking



Experiment Roster (df_final_experiment_clients)

- Control vs Test group assignment

Data Preparation

Dataset Integration

Merged multiple datasets into a single unified structure for analysis

Data Filtering

Excluded users without a valid starting point to ensure complete session tracking

Data Cleaning

Removed missing values and outliers to ensure data consistency and accuracy

Experiment Segmentation

Split users into Control and Test groups based on experiment assignment

User Journey Structuring

Reconstructed user journeys using session IDs and timestamps to analyze step transitions

Clients Demographics

Who Are Vanguard's Online Clients?

Metric	**Test Group**	**Control Group**
Age	47	47
Client Tenure (years)	11.98	12.09
Number of Accounts	2.25	2.26
Total Balance ($)	128,962.61	150,151.90
Calls (last 6 months)	3.06	3.13
Logins (last 6 months)	6.1	6.17

Client Behavior & Initial Insights

Moderate platform usage → Most clients logged in 3–9 times in 6 months

Low call center interaction → Majority had few support calls

Indicates → Users are digitally active and relatively independent

User journey follows a funnel structure → Drop-offs occur between steps

Repeated steps observed → Suggests confusion or friction

Variation in time spent across steps → Some steps are more complex or unclear

The KPIs

Completion Time

- Average time spent per step

Completion Rate

- % of users progressing to the next step

Error Rate

- % of users going backward in the process

Drop-Off Rate

- Users who live the process berofe completing it

HYPOTHESIS TESTING

Completion Time

For each step:

- H_0 (Null Hypothesis): The mean completion time is equal between Test and Control groups
- H_1 (Alternative Hypothesis): The mean completion time differs between Test and Control groups
- **Output**: Reject null hypothesis (p-value < 0.001)

Completion Rate

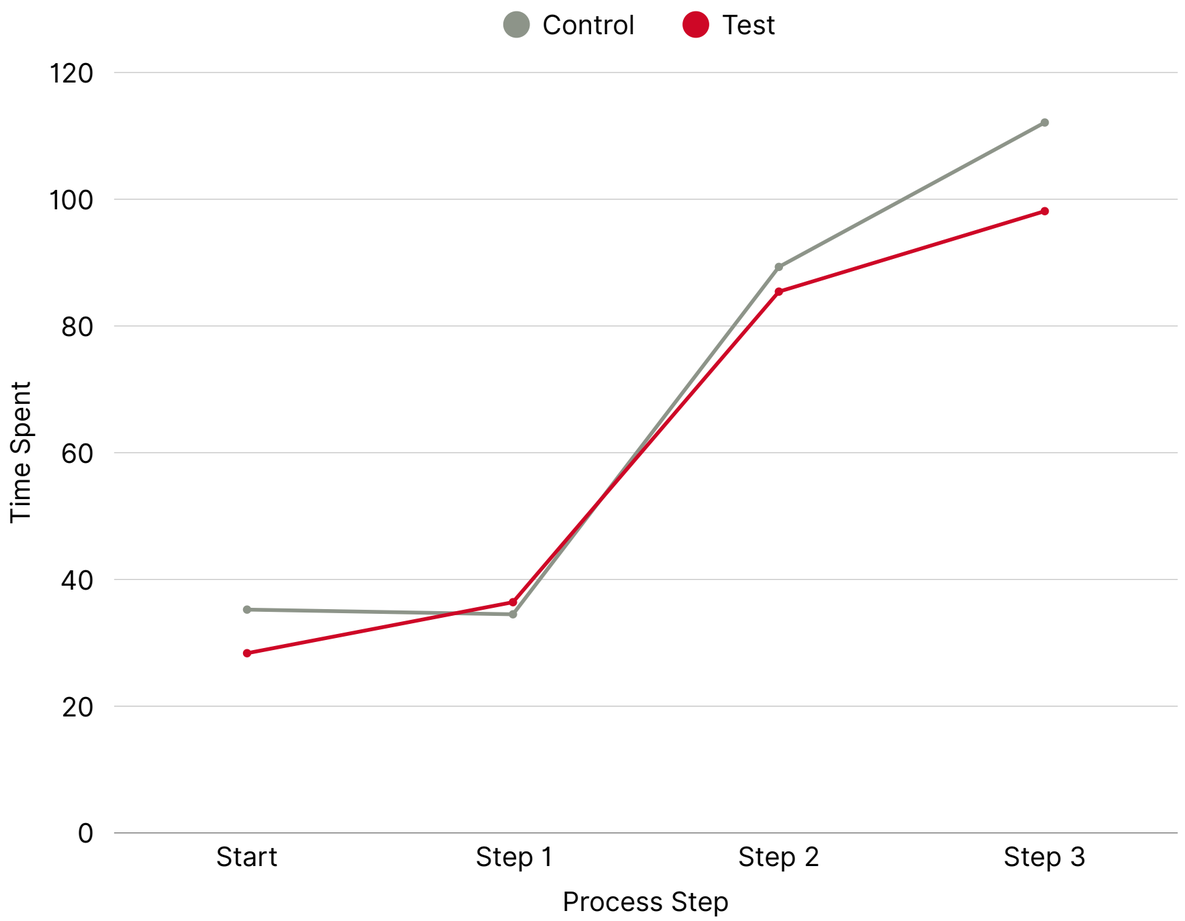
- H_0 (Null Hypothesis): The test completion rate is less than or equal to the control completion rate
- H_1 (Alternative Hypothesis): The test completion rate is greater than the control completion rate
- **Output**: Reject null hypothesis (p-value > 1.96)

Error Rate

- H_0 (Null Hypothesis): The error rate is equal between Test and Control groups
- H_1 (Alternative Hypothesis): The error rate differs between Test and Control groups
- **Output**:

All tests were conducted at a 5% significance level ($\alpha = 0.05$).

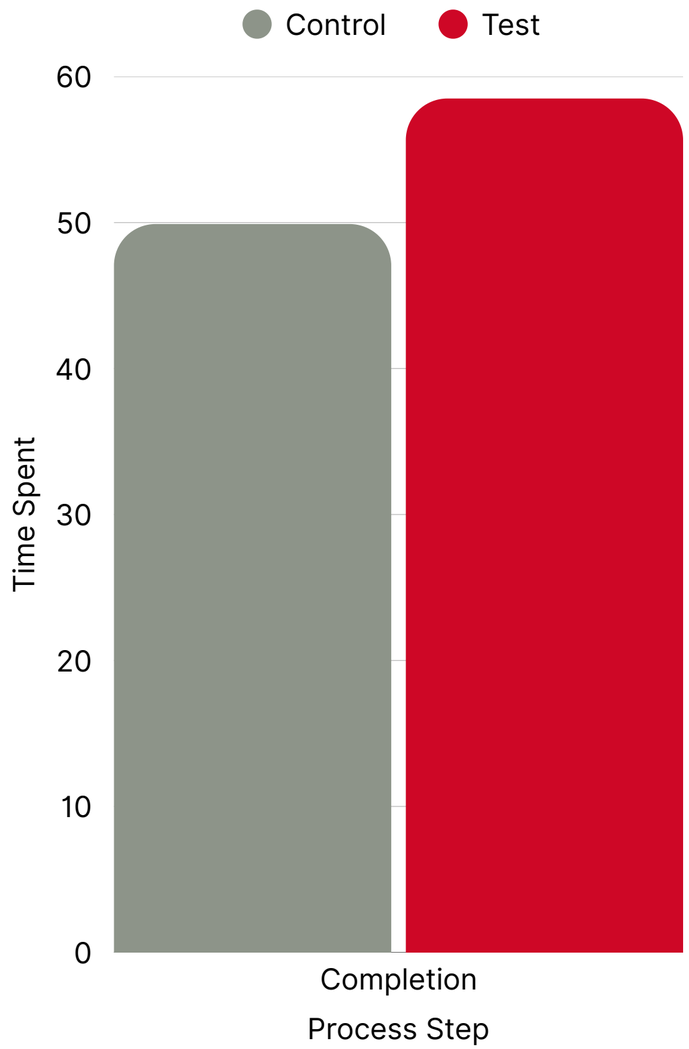
Completion time



- ANOVA - Compare step times within groups
- Two-Sample T-Test - Compare step times between groups

Step	Control	Test	Δ
Start	35.24	28.36	Faster
Step 1	34.50	36.41	Slower
Step 2	89.32	85.42	Faster
Step 3	112.11	98.12	Faster

Completion rate



The proportion of users who reach the final 'confirm' step.

Control: 49.9%
Test: 58.5%

+8.7pp ↑

Insight: New web design meets Vanguard's minimum increase in completion rate by at least **5%**

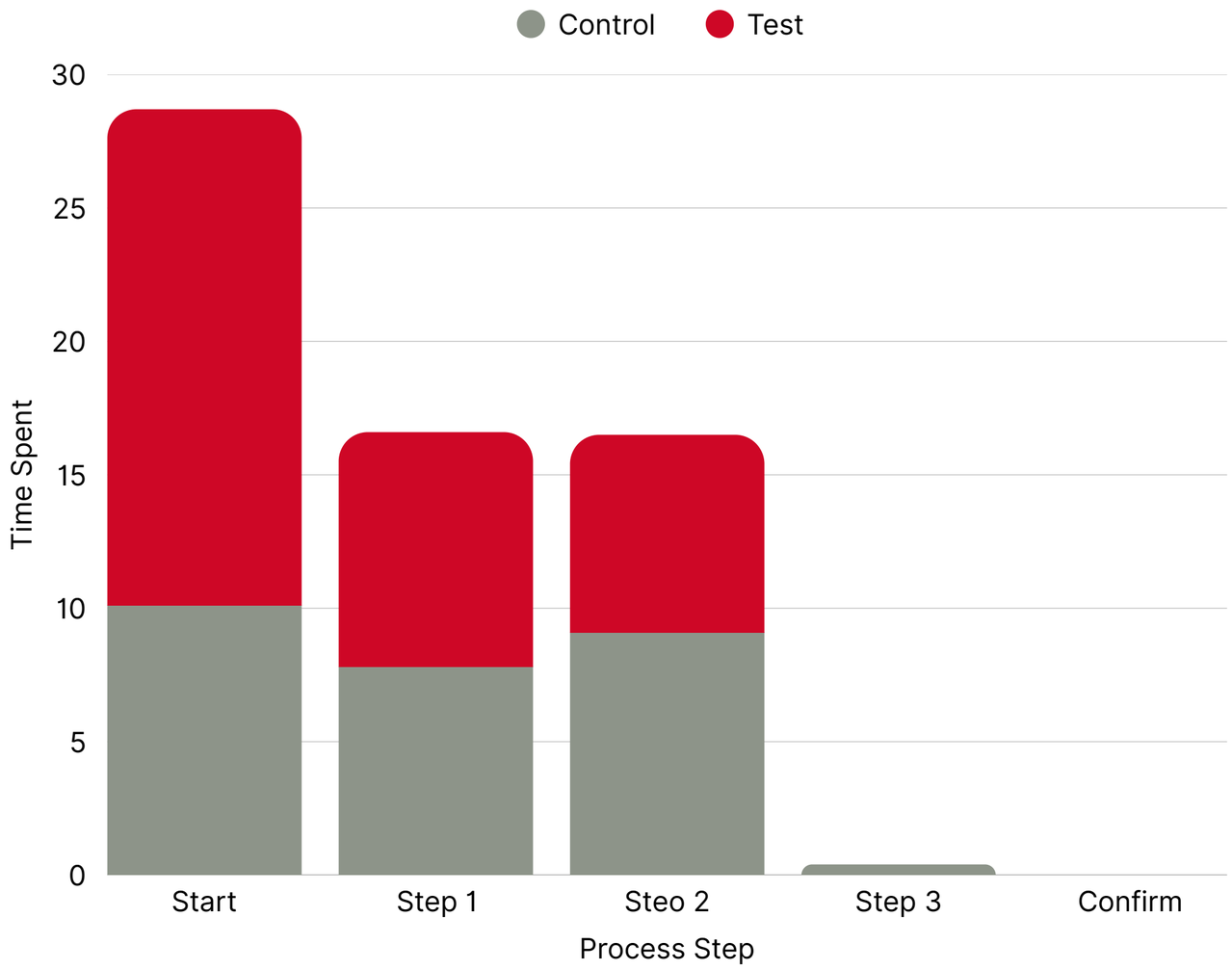
Drop off rate

Key Insights

- Drop-off improved significantly in first steps: new interface helps users move past the first stage more easily.
- Later steps → higher drop-off in the test version: potential friction introduced in the final stages.
- Early-stage improvements → biggest impact.
- There's a trade-off → new design improves early funnel progression (refinement in final steps required)

Step	Control	Test	Δ
Start	0	0	Same
Step 1	25.8	15.5	Lower
Step 2	15.7	14	Lower
Step 3	10.3	10.6	Higher
Confirm	16.7	17.5	Higher

⚠ Error rate



Error Rate (Overall)

Control: 20.3%
Test: 26.8%

+6.5pp ↑

Insight: The new UI introduces more friction early in the journey, leading to a significant increase in user errors.

Step	Control	Test	Δ
Start	10.1	18.6	Higher
Step 1	7.8	8.8	Higher
Step 2	9.1	7.4	Lower
Step 3	0.4	0.0	Lower
Confirm	0.0	0.0	Same

EXPERIMENT EVALUATION

Experimental Design

Randomization

- Clients were split into Test and Control groups
- Groups show similar demographics and behavior → Indicates successful randomization

Duration

- Experiment ran for ~3 months
- Sufficient to capture:
  - Different user sessions
  - Behavioral patterns over time

Potential Biases & Limitations

Repeated Visits

- Same user may appear multiple times

→ Could bias results if not properly aggregated

User Familiarity

- Some users more experienced (logins, tenure)

→ May influence performance independently of UI

Error Definition

- Errors inferred from backward steps

→ May not capture all usability issues

EXPERIMENT EVALUATION

What Could Improve the Analysis?

Additional Data Needed

- Explicit error tracking → Direct measure of user mistakes
- User intent / task success → Did users achieve their goal?
- Session-level context → Device, browser, entry point
- User feedback (qualitative) → Understand why users struggle

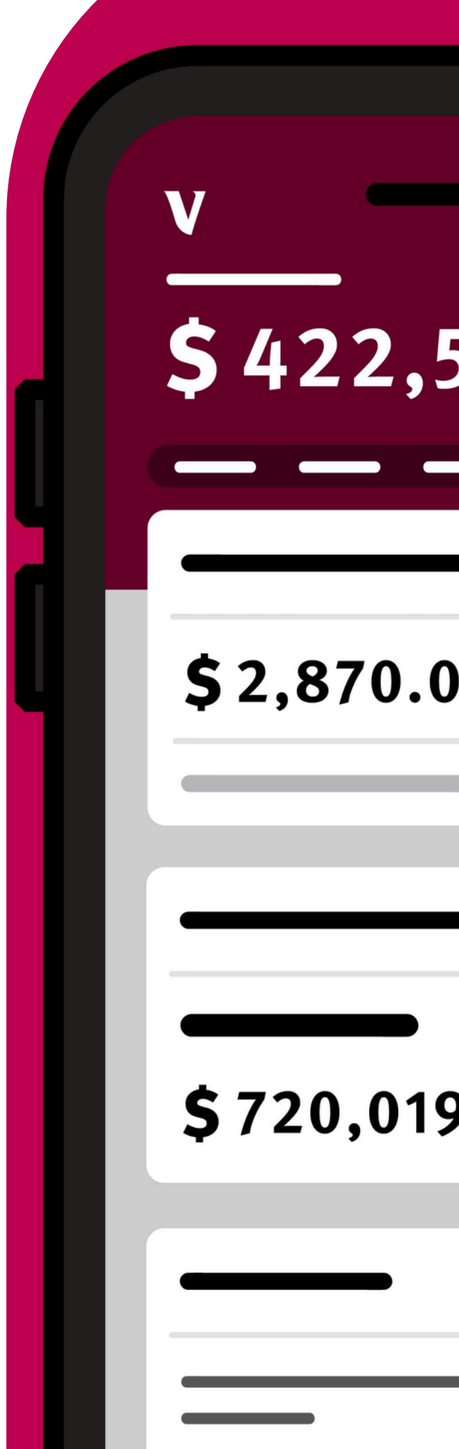
DASHBOARD

https://public.tableau.com/app/profile/victoria.cano.acha/viz/VANGUARDDASHBOARD/Dashboard1?publish=yes

CONCLUSIONS

Final Verdict: Adopt the new UI with targeted improvements The new UI is more effective at driving users forward and improving efficiency, but introduces usability challenges at the beginning and end of the journey

1. Higher number of users who reached “confirmed” step
2. Higher conversion → Stronger early engagement
3. Faster user journey → Biggest gains in Start and Step 3
4. Higher error rate → Concentrated at the start of the process
5. Mixed drop-off performance → Lower drop-off in early steps and slight increase near completion

Recommendations:

- Improve onboarding clarity (Start & Step 1)
- Reduce early-stage errors
- Optimize final steps to reduce drop-off